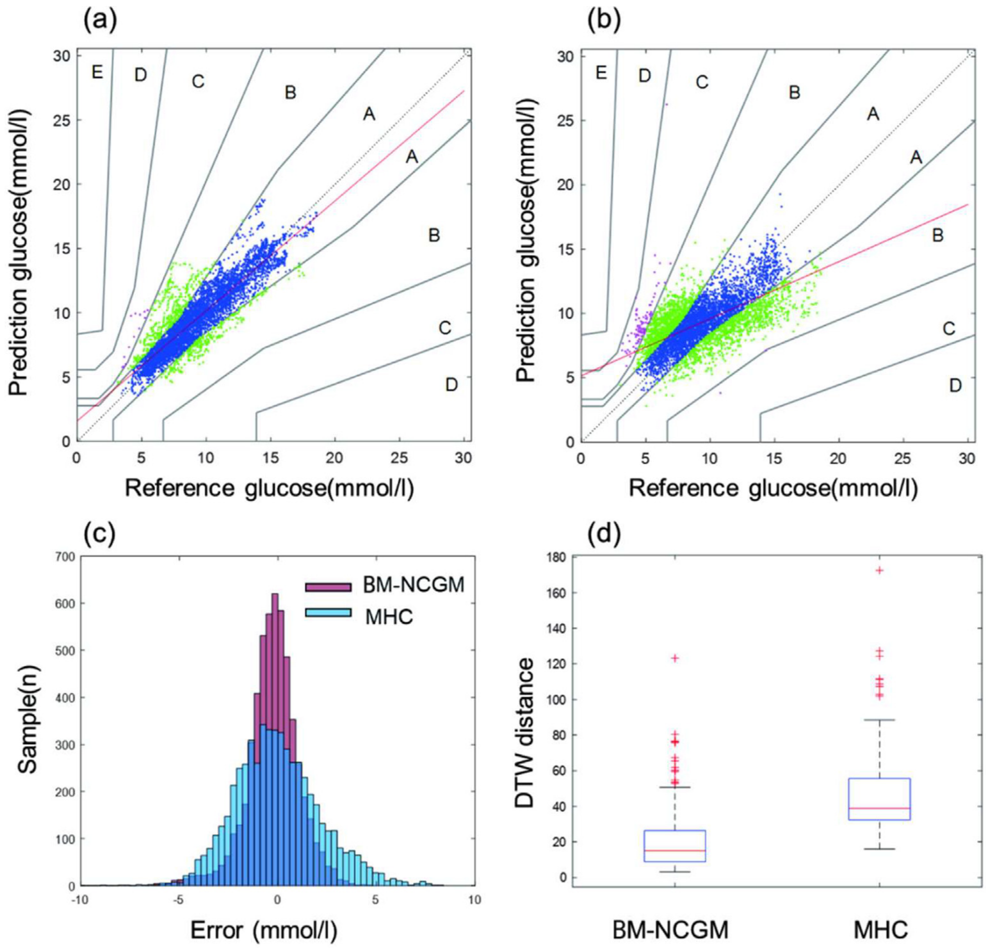

Fig : Performance comparison of the two models: (a) BM-NCGM prediction results, (b) blood glucose prediction results of MHC heat model, (c) error distribution of the two blood glucose prediction models, and (d) DTW analysis of the two blood glucose prediction models

accounted for 77.58 %, and the A+B area for 99.57 %. The correlation coefficient (0.85), RMSE (1.48 mmol/L) and MARD (11.51 %) showed an improvement of over 32 % compared to the non-use of BM-NCGM. The Dynamic Time Warping algorithm was used to calculate the distance between the predicted blood glucose spectrum and the reference blood glucose spectrum, with an average distance of 21.80, demonstrating the excellent blood glucose spectrum tracking ability of BM-NCGM.

Conclusion: This study is the first to apply the Bergman minimum model to non-invasive continuous blood glucose monitoring research, supported by a large amount of clinical trial data, bringing non-invasive continuous blood glucose monitoring closer to its true application in daily blood glucose monitoring.

https://doi.org/10.1016/j.diabres.2025.112805

BA2025-0324

Performance and accuracy of ten AI-based algorithms in diabetic retinopathy screening in real-world clinical setting

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Background: Diabetic retinopathy (DR) is a common complication of diabetes and may lead to irreversible visual loss. Efficient screening and improved treatment of both diabetes and DR have amended visual prognosis for DR. Unfortunately, screening is not always available due to limited access to expert services. Artificial intelligence (AI) based analysis have been suggested to help with the resource shortage.

Aim: We studied the performance of twenty-five AI algorithms to assess if they could be used as an aid in the screening workflow.

Method: The image collection and the concept of algorithm comparison are described in articles by Kubin and colleagues (2021, 2024). Here, we focus on the ten algorithms that had the